

## **IT580-Advanced Algorithms (3-0-2-4)**

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**Objectives:** The course aims to enhance students' skills in problem solving techniques, understanding and analysing of algorithms. Emphasis will be given on problem solving approaches using divide-and-conquer, greedy, dynamic programming, randomization, and approximation algorithms.

**Outcomes:** After completion of the course, the students will be able to understand different problem solving approaches surrounded in various contexts and real-world applications.

### **Books:**

Introduction to Algorithms -- Cormen, Leiserson, Rivest, Stein (MIT Press)  
Algorithms -- Dasgupta, Papadimitriou, and Vazirani (McGraw-Hill Education).

### **Grading Policy:**

Class participation/Scribe/Presentation – 20%  
Lab practice and performance – 20%  
Mid-semester exam – 20%  
End Semester Exam – 40%.